

**Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering, Pune**



B.Tech Final Year Project

CipherDocs – A secure, AI-powered solution to prevent forgery of critical & official documents using encryption and blockchain-based verification.

Group No: B18

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Abstract:

This research presents the design and implementation of a decentralized system for tackling document forgery leveraging blockchain technology (Polygon/Ethereum) integrated with artificial intelligence to enhance usability, security, and authentication. The system employs smart contracts for immutable and transparent management of file metadata, facilitating secure and automated control over file operations such as upload, retrieval, and expiry-based deletion. Files are distributed and stored across a decentralized network, ensuring high availability and fault tolerance. Dynamic QR codes generated within the system provide seamless, secure access to files, while AI-driven validation mechanisms authenticate user requests and verify data integrity. This fusion of blockchain with AI creates a robust ecosystem that addresses challenges of centralized storage systems by offering transparency, censorship resistance, and efficient file lifecycle management. The proposed approach promotes scalable, user-friendly decentralized data storage suitable for a broad range of general-purpose applications.

Keywords: Artificial Intelligence, Blockchain, Smart Contracts, Fraud Prevention, Encryption, Document Verification

Introduction:

Document forgery, tampering, and unauthorized access pose critical challenges in secure document storage and authentication. Centralized systems suffer from data breaches and lack transparency, reducing trust and security. Decentralized file storage with blockchain (Polygon or Ethereum) offers immutable, transparent data management that prevents document alteration and forgery. When combined with AI, the system enables intelligent authentication and dynamic validation. Our project aims to build a decentralized storage platform leveraging blockchain and AI for easy connection, authentication, and file retrieval. Features include expiry-based storage, dynamic QR codes, and AI-powered validation, creating a secure and scalable solution that prevents forgery and enhances document management.

Problem Statement:

Despite recent progress in blockchain-based document verification, several challenges still persist. Public blockchains such as Ethereum often face high transaction costs and scalability issues, making large-scale deployment difficult. Existing forgery detection models are limited in their capabilities, as they lack transparency and often focus on a single modality, which reduces their ability to detect complex tampering involving both text and images. Traditional watermarking and encryption techniques remain fragile, as they can break during file conversions or when faced with

sophisticated AI-generated forgeries. Moreover, verification processes in current systems still involve manual steps, which slows down adoption and creates inefficiencies. Privacy is another critical concern, since issuers and verifiers frequently gain access to more data than is actually necessary for validation. To address these gaps, CipherDocs proposes a fully automated solution that integrates AI with blockchain, enabling secure issuance, validation, and verification of documents. By streamlining these processes, the system minimizes manual effort while enhancing trust, scalability, and privacy.

Objectives:

- To design a tamper-proof certificate issuance and verification platform using Ethereum/Polygon blockchain.
- To integrate agentic AI workflows that automate encryption, forgery detection, document summarization, and verification.
- To implement advanced verification features such as dynamic QR codes, Zero-Knowledge Proofs, and automated revocation/expiry.
- To ensure seamless usability for issuers, students, and verifiers with chatbot assistance and intuitive interfaces.

Methodology:

1. System Workflow

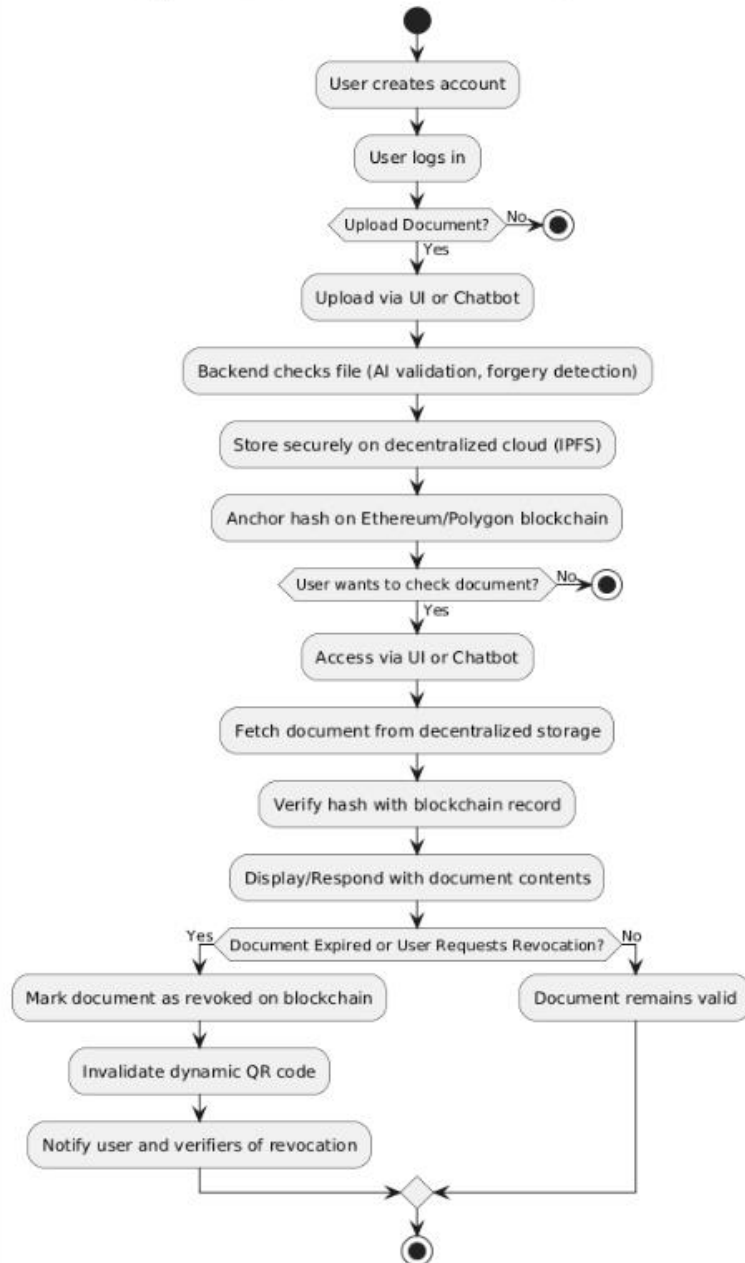
- Issuer: Creates a digital certificate, which is hashed, signed, and anchored on blockchain. AI decides encryption levels per section. A dynamic QR code is generated and embedded.
- Student: Accesses, downloads, and shares their certificate securely anytime. Expiry/revocation is managed automatically.
- Verifier: Scans QR code or enters ID → system fetches blockchain-anchored data → AI validates integrity, checks for tampering, and provides instant authenticity results.

2. Technologies Used

- Blockchain Layer: Ethereum/Polygon for certificate anchoring.
- Storage: Cloud based decentralized document storage.
- AI Modules:
 - Forgery detection (multimodal CNN/Transformer models).
 - Context-aware encryption (AI decides sensitivity levels).
 - Summarization/Review (LLMs for student-facing summaries).

- Agentic AI orchestration (automated workflows for issuance/verification).
- Verification Features: Dynamic QR codes, Zero-Knowledge Proofs, revocation/expiry smart contracts.
- Access Control: Secure login via MetaMask or institutional wallets.

CipherDocs Document Flow with Revocation



Results:

- Tamper-proof certificates issued and stored on blockchain.
- Dynamic encryption ensures section-wise data protection.
- Forgery-resistant verification through AI-powered multimodal anomaly detection.
- Privacy-preserving proofs with ZKPs.
- Ease of use: students get instant access, verifiers confirm in seconds, issuers eliminate manual checks.
- Explainable AI outputs highlight tampered regions with reasons, increasing trust in verification.

Applications / Significance

- Education: Degree/diploma verification by universities.
- Government: Issuance of licenses, ID cards, legal documents.
- Finance: Verification of contracts, financial statements.
- Healthcare: Preventing forgery of medical records and prescriptions.
- Corporate: Employment verification, KYC/AML compliance.

By integrating agentic AI + blockchain, CipherDocs reduces fraud, improves trust, and streamlines processes for millions of users. It directly supports SDGs like Reduced Inequalities (SDG 10), Peace, Justice & Strong Institutions (SDG 16), and Industry, Innovation, and Infrastructure (SDG 9).

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